SYLLABUS : HEAT TRANSFER & ALTERNATING CURRENT

1. The current in a circuit varies with time as $I = 2\sqrt{t}$. Then the rms value of the current for the interval t = 2 to t = 4 sec is

(A) $\sqrt{3}A$ (B) $2\sqrt{3}A$ (C) $\sqrt{3}/2A$ (D) $(4-2\sqrt{2})$ A

2. In ac circuit when ac ammeter is connected it reads i current if a student uses dc ammeter in plane of ac ammeter the reading in the dc ammeter will be

(A) $\frac{i}{\sqrt{2}}$ (B) $\sqrt{2}i$ (C) 0.637 i (D) zero

3. An AC current is given by $I = I_0 + I_1 \sin \omega t$ then its rms value will be

(A) $\sqrt{I_0^2 + 0.5I_1^2}$ (B) $\sqrt{I_0^2 + 0.5I_0^2}$ (C) 0 (D) $\frac{I_0}{\sqrt{2}}$

4. The effective value of current $i = 2 \sin 100\pi t + 2 \sin (100\pi t + 30°)$ is

(A) $\sqrt{2}A$ (B) $2\sqrt{2+\sqrt{3}}$ (C) 4 (D) None

5. r.m.s. value of current $i = 3 + 4 \sin (\omega t + \pi/3)$ is:

(A) 5 A (B) $\sqrt{17}$ A (C) $\frac{5}{\sqrt{2}}$ A (D) $\frac{7}{\sqrt{2}}$ A

6. The peak value of an alternating e.m.f given by $E = E_0 \cos \omega t$, is 10 volt and frequency is 50 Hz. At time t = (1/600) sec, the instantaneous value of e.m.f is :

(A) 10 volt (B) $5\sqrt{3}$ volt (C) 5 volt (D) 1 volt

7. The voltage of an AC source varies with time according to the equation, $V = 100 \sin 100 \pi t \cos 100 \pi t$. Where t is in second and V is in volt. Then :

(A) the peak voltage of the source is 100 volt

(B) the peak voltage of the source is $(100/\sqrt{2})$ volt

(C) the peak voltage of the source is 50 volt

(D) the frequency of the source is 50 Hz

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8. An alternating voltage is given by : $e = e_1 \sin\omega t + e_2 \cos\omega t$. Then the root mean square value of voltage is given by :

(A) $\sqrt{e_1^2 + e_2^2}$ (B) $\sqrt{e_1 e_2}$ (C) $\sqrt{\frac{e_1 e_2}{2}}$ (D) $\sqrt{\frac{e_1^2 + e_2^2}{2}}$

9. In an AC circuit the voltage applied is $E = E_0 \sin\omega t$. The resulting current in the circuit is $I = I_0 \sin\left(\omega t - \frac{\pi}{2}\right)$. The power consumption in the circuit is given by :–

(A) $P = \frac{E_0 I_0}{\sqrt{2}}$ (B) P = zero (C) $P = \frac{E_0 I_0}{2}$ (D) $P = \sqrt{2}\, E_0 I_0$

10. The potential difference V across and the current I flowing through an instrument in an AC circuit are given by :

$V = 5 \cos\omega t$ volt

$I = 2 \sin\omega t$ Amp.

The power dissipated in the instrument is :

(A) zero (B) 5 watt (C) 10 watt (D) 2.5 watt

11. A direct current of 2 A and an alternating current having a maximum value of 2 A flow through two identical resistances. The ratio of heat produced in the two resistances in the same time interval will be:

(A) 1 : 1 (B) 1 : 2 (C) 2 : 1 (D) 4 : 1

12. A sinusoidal AC current flows through a resistor of resistance R. If the peak current is I_p, then average power dissipated is :

(A) $I_p^2 R\cos\theta$ (B) $\frac{1}{2} I_p^2 R$ (C) $\frac{4}{\pi} I_p^2 R$ (D) $\frac{1}{\pi^2} I_p^2 R$

13. What is the rms value of an alternating current which when passed through a resistor produces heat, which is thrice that produced by a D.C. current of 2 ampere in the same resistor in the same time interval?

(A) 6 ampere (B) 2 ampere (C) $2\sqrt{3}$ ampere (D) 0.65 ampere

14. A resistor and a capacitor are connected to an AC supply of 200 volt, 50 Hz in series. The current in the circuit is 2 ampere. If the power consumed in the circuit is 100 watt, then the resistance in the circuit is:

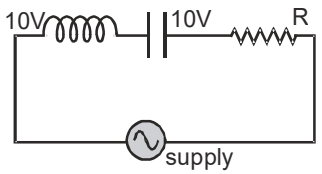
(A) 100 Ω (B) 25 Ω (C) $\sqrt{125 \times 75}$ Ω (D) 400 Ω

15. The impedance of a series circuit consists of 3 ohm resistance and 4 ohm reactance. The power factor of the circuit is :

(A) 0.4 (B) 0.6 (C) 0.8 (D) 1.0

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16. If value of R is changed, then :–



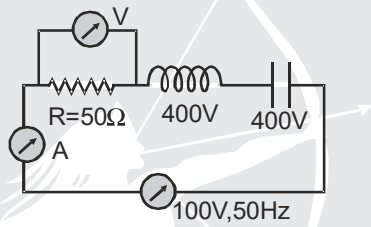
(A) voltage across L remains same

(B) voltage across C remains same

(C) voltage across LC combination remains same

(D) voltage across LC combination changes

17. In given LCR circuit, the voltage across the terminals of a resistance and current will be :–



(A) 400V, 2A (B) 800V, 2A (C) 100V, 2A (D) 100V, 4A

18. A 0.21-H inductor and a 88-Ω resistor are connected in series to a 220-V, 50-Hz AC source. The current in the circuit and the phase angle between the current and the source voltage are respectively. (Use $\pi = 22/7$)

(A) 2 A, $\tan^{-1} 3/4$ (B) 14.4 A, $\tan^{-1} 7/8$ (C) 14.4 A, $\tan^{-1} 8/7$ (D) 3.28 A, $\tan^{-1} 2/11$

19. A 100 volt AC source of angular frequency 500 rad/s is connected to a LCR circuit with L = 0.8 H, C = 5 μF and R = 10 Ω, all connected in series. The potential difference across the resistance is

(A) $\frac{100}{\sqrt{2}}$ volt (B) 100 volt (C) 50 volt (D) $50\sqrt{3}$

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20. In an AC circuit, a resistance of R ohm is connected in series with an inductance L. If phase angle between voltage and current be 45°, the value of inductive reactance will be.

(A) R/4 (B) R/2

(C) R (D) cannot be found with the given data

21. In an AC circuit the potential differences across an inductor and resistor joined in series are respectively 16 V and 20 V. The total potential difference across the circuit is

(A) 20 V (B) 25.6 V (C) 31.9 V (D) 53.5 V

22. An AC voltage source V = $200\sqrt{2}$ sin 100 t is connected across a circuit containing an AC ammeter(it reads rms value) and capacitor of capacity 1 μF. The reading of ammeter is :

(A) 10 mA (B) 20 mA (C) 40 mA (D) 80 mA

23. In an LCR circuit, capacitance is changed from C to 2C. For the resonant frequency to remain unchanged, the inductance should be changed from L to–

(A) 4 L (B) 2L (C) L/2 (D) L/4

24. In a series resonant LCR circuit, the voltage across R is 100 volts and R = 1 kΩ with C = 2μF. The resonant frequency ω is 200 rad/s. At resonance the voltage across L is–

(A) 2.5×10^{-2} V (B) 40 V (C) 250 V (D) 4×10^{-3} V

25. A power transformer (step up) with an 1 : 8 turn ratio has 60 Hz, 120 V across the primary; the load in the secondary is 10^4 Ω. The current in the secondary is

(A) 96 A (B) 0.96 A (C) 9.6 A (D) 96 mA

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ANSWER KEY

1.	(B)	**2.**	(D)	**3.**	(A)	**4.**	(D)	**5.**	(B)
6.	(B)	**7.**	(C)	**8.**	(D)	**9.**	(B)	**10.**	(A)
11.	(C)	**12.**	(B)	**13.**	(C)	**14.**	(B)	**15.**	(B)
16.	(C)	**17.**	(C)	**18.**	(A)	**19.**	(B)	**20.**	(C)
21.	(B)	**22.**	(B)	**23.**	(C)	**24.**	(C)	**25.**	(D)